Stage 5 — Check Your Understanding

Q1. Repairing the flow next to the fibre

Blob width	Slip at r = a
0.1a	0.0584U = 5.84% of U
0.05a	0.0311U = 3.11% of U

Halving the blob width decreases the slip by (0.0584 – 0.0311)/0.0584 × 100 = **46.75%**, consistent with first-order error O(epsilon). Capture depends on the wall shear and flow within a few hundredths of a fibre radius, so even 3% false surface velocity can alter grazing trajectories. Permeability instead averages transport and drag across the cell and is far less sensitive to this local slip.

Proposition 9 reconstructs an exact annular Stokes flow on a < r < 1.3a, imposing zero velocity at r = a and matching the outer flow. The reported near-wall error improves from 8.6×10^{-3}U to 3.6×10^{-4}U. The draft identifies the measurement location as the physical fibre surface r = a; the exact code statement in scripts/stage5_validation.py, Section 2, is not available for direct verification.

Q2. Brownian-bridge correction

Given d0 = d1 = 0.01a, D = 0.02, h = 0.01:

$\sigma^2 = 2Dh = 2(0.02)(0.01) = 0.0004$.

$p = \exp(-2d_0d_1/\sigma^2) = \exp[-2(0.01)(0.01)/0.0004] = \exp(-0.5) =$ **0.60653 (60.65%)**.

Step h	With bridge	Without bridge
0.04	0.0565	0.0548
0.02	0.0572	0.0556
0.01	0.0574	0.0564
0.005	0.0585	0.0569

The bridge formula assumes a locally flat absorbing wall and constant normal diffusivity over the step. This is justified when displacement is small compared with fibre radius. Without the correction, trajectories that touch and return between recorded endpoints are missed, systematically lowering capture.

Q3. Inertia below the impaction range

Stokes number	Loss vs exact interception
0.0116	3.6%
0.0465	11.5%
0.1862	20.6%

Sketch in words: air moves from left to right and curves around a circular fibre; mark a grazing particle near the upper or lower shoulder, where the streamline bends sharply, and a second particle approaching the front stagnation region. Inertia carries grazing particles outward rather than around the curved shoulder, reducing interception, but carries

upstream particles into the front, increasing impaction. In this range the shoulder loss dominates.

With diffusion, inertia changes capture from 0.01477 to 0.01223: (0.01477 – 0.01223)/0.01477 × 100 = **17.20%**. The kinetic boundary length is lambda = sqrt(D St) = 0.0070a (reported). Pulling the absorbing circle inward by 1.03 lambda and 1.5 lambda reduces capture by 5.6% and 8.9%, respectively. The kinetic layer explains roughly half of the 17% loss; outward shoulder drift explains the rest.

Q4. From one cell to a whole filter

f = 0.0122, alpha = 0.1, Lx = 5.605, df = 2.

> –ln(1 – f) = –ln(0.9878) = 0.012275.
>
> eta = –ln(1 – f) × pi df/(4 alpha Lx) = 0.012275 × (2 pi)/(4 × 0.1 × 5.605) = **0.03440**.
>
> For L/df = 166.7: P = exp[–4 alpha eta (L/df)/pi] = exp[–4(0.1)(0.03440)(166.7)/pi] = **0.4818** ≈ **0.482**.

The logarithm arises because independent cell penetrations multiply: P = (1 – f)^(L/Lx). Hinds' Stage 1 convention adds 1/(1 – alpha) = 1/0.9 = 1.1111 to the attenuation exponent by crediting each fibre with the interstitial rather than face velocity. This produces the two classical Table 8 penetrations: 0.357 (Hinds) versus 0.396 (Eq. 8.6).

Q5. Statistics

Random cells are resampled as complete geometric realizations because between-cell structure variation is the dominant uncertainty. Regular arrays have fixed geometry, so finite capture counts are redrawn binomially. A random/hexagonal ratio must propagate uncertainty in both numerator and denominator; otherwise its interval is too narrow.

Random identical/hexagonal capture ratio = **0.942 [0.863, 1.075]** (95% interval). Random CV = 0.5 / random identical capture ratio = **0.822 [0.635, 1.053]**. Both intervals contain 1, so neither comparison establishes a capture difference. Six independent random cells make the estimate of between-cell variation imprecise and limit conclusions about fibre-size spread. "Not shown" is not equivalent to "no effect."